PNG University of Technology

School of Mathematics & Computer Science

Project Specification (v4) for "Purchasing and Inventory tracking" system

[Original version1: July 2026]

[Revised version4: 18 August 2026]

1 Project Background

The School of MCS purchases many items to support its operations during an academic year. Some purchased items are immediately used while other items are held as parts or inventory and used later when needed. The school is also required to account for all its expenditure through regular reporting. The purchased items include:

- Assets – Computers, ICT Equipment, Printers, Tools & Equipment, Furniture & Fixtures
- Consumables - Stationery, Cleaning, General, Staff-use

The school currently manually handles purchasing activities, inventory management, and reporting. The proposed system should provide a platform for the school to efficiently manage purchasing and inventory while maintaining data integrity and producing reports to comply with management reporting obligations.

2 Project Objective

The main objective of the project is to:

- Develop a complete database-driven application that integrates - ***Purchasing and Inventory*** - into an information management system.

The proposed system should be able to assist the school perform these specific tasks:

- Manage entities -> Suppliers, Assets, Consumables, Inventory, Customers
- Manage processes -> GE requests, Purchases, Inventory movements
- Produce reports -> Management reports, Operational reports

3 Project Scope and Inventory Flag

The school currently processes GE orders to facilitate payments on all types of goods and services incurred by the school. The scope covers purchases of both GOODs and SERVICEs. Goods that are used regularly to support the operations and upkeep of the school are to be converted into inventory and referred to as STOCK items. Other expenditure items such as SERVICES and/or ONE-OFF payments are referred to as NON-STOCK items and are not recorded and/or tracked in the inventory.

A major processing option ("Inventory Flag") on a GE order will determine whether the order is requested for STOCK or NON-STOCK items. The option is flagged during creation of GE orders:

- STOCK -> identifies an order as a STOCK order. Purchased goods are posted and tracked in inventory.
- NON-STOCK -> identifies an order as a NON-STOCK order. Purchased items are not recorded in inventory.

4 Functional Modules

4.1 Purchasing

Features

4.1.1 GE Order (Purchase Request)

A **General Expenditure (GE)** Order is the purchase request initiated by staff at MCS. A typical purchasing cycle is as follows:

1. A GE order is generated as per supplier quote.
2. Approved GE is converted into a Purchase Order and forwarded to the supplier.
3. The supplier fulfills the request as per the GE order and MCS pays from the school's account.
4. The GE order is complete once MCS receives supplier's invoice.

MCS staff will have full access to Create/Edit/Cancel/Delete and view all GE orders.

Each "***ORDERS***" table contains the following attributes:

Attribute	Data type	Explanation	Extra Note
GE Number	Integer [PK]	Unique GE number of purchase request.	Pre-allocated GE number
Inventory Flag	Text	Flag to control inventory interface	Allowed values {STOCK, NON-STOCK}
PO Number	Integer [FK]	Unique PO number linked to GE	Pre-allocated PO number. Defaults to GE number
SupplierID	Integer [FK]	ID of supplier	Validate SUPPLIER
Date	Date	Creation date. Defaults to date of entry	System default
UserID	Text [FK]	Login ID of the user	System default
BranchID	Integer [FK]	ID of the ordering branch.	Defaults to '201'
Account Code	Text	Accounting code for reporting	
ItemID	Integer [FK]	ID of the stock item. Null for NON-STOCK item	Validate ITEM table [for STOCK order]
Item	Text	Description (Particulars) of line item	Get from ITEM [for STOCK order]
UOM	Text	UOM of the item	Get from ITEM [for STOCK order]
Quantity	Integer	Units to order	
Unit Cost	Decimal	Buying price per unit	
Total Cost	Decimal	Extended buying price	[Quantity * Unit Cost].

Status	Text	{Pending} →{Approved/Rejected/Cancelled} → {Ordered} →{Received} →{Backorder/Complete}	Defaults to 'Pending'. Manually updated.

4.1.2 Supplier

Supplier is an entity that provides goods and services to the school to support its operations. Suppliers are mostly PNG-based with respective payment terms.

MCS staff will have full access to Create/Edit/Delete and view all suppliers.

Each "***SUPPLIER***" table contains the following attributes:

Attribute	Data type	Explanation
SupplierID	Integer [PK]	Unique ID of supplier. [4-digit range: 5001-5999]
Supplier	Text	Name of supplier
Address	Text	Full address of supplier
Contact	Text	Name and details of contact person
Payment Term	Text	Allowed values {CASH, CREDIT}
Currency	Text	Payment currency. Defaults to 'PGK'

4.1.3 Purchase Receipts

Once the requested goods/services are consumed or received on-site at MCS, the corresponding PO (or GE) is received into the system to complete the PO cycle. Receipt of a STOCK order automatically updates inventory balances of the respective line items.

MCS staff will have full access to Create/Edit/Cancel/Delete and view all receipted orders.

Each "***RECEIPT***" table contains the following attributes.

Attribute	Data type	Explanation	Extra Note
PO Number	Integer [PK]	PO (or GE) number of the GE order.	Validate ORDERS
BranchID	Integer [FK]	ID of receiving branch. Defaults to '201'	Get from ORDERS [view only]
ItemID	Integer [FK]	ID of ordered item. Nulls for non-stock order	Get from ORDERS [view only]
SupplierID	Integer [FK]	ID of the supplier	Get from ORDERS [view only]
Quantity Ordered	Integer	Quantity on order	Get from ORDERS [view only]
Quantity Received	Integer	Physical units received at MCS	Allow partial receipt
UOM	Text	Unit of Measure	Get from ORDERS [view only]

Received By	Text	Name of MCS staff	
Date Received	Date	Date goods received	
GRN	Text	Goods receipt number [i.e. invoice number]	Provided by the supplier
Account Code	Text	Accounting code for reporting	Get from ORDERS [view only]

Upon successful receipt of a STOCK order, the system updates the following parameters in ITEMBRANCH by *ItemID/BranchID* for each line item:

1. increases **Current Stock** level in the receiving branch
2. re-calculates **Unit Cost** of the item in the receiving branch using formula below:

$$\boldsymbol{Unit\ Cost} = \frac{\boldsymbol{AmountReceived + currentValue}}{\boldsymbol{QtyRecieved + currentStock}}$$ where

AmountReceived -> value of quantity received
QtyReceived -> number of units received
currentStock -> Current Stock in the receiving branch before PO receipt
currentValue -> current value [Current Stock * Unit Cost] of the item in the receiving branch before PO receipt

4.2 Inventory Management

Features

4.2.1 Item

Items purchased by the school to support regular upkeep and operations of the school are catalogued as STOCK items.

MCS staff will have full access to Create/Edit/Delete and view all stock items.

Each "***ITEM***" table contains the following attributes:

Attribute	Data type	Explanation	Extra Note
ItemID	Integer [PK]	Unique ID of stock item.	[4-digit range: 1001-1999]
Item	Text	Full name or description of stock item	
UOM	Text	Unit of Measure (or packaging)	
Category	Text	Product category	ASSET or CONSUMABLE
Sub-Category	Text	Product sub-category	Asset -> {Computer, ICT Equipment, Tools Equipment, Fixtures & Furniture} Consumable -> {Stationery, Cleaning, General, Staff}
SupplierID	Integer [FK]	ID of primary supplier (or Manufacturer)	

4.2.2 Branch

Branch is a physical facility or site that facilitates functional operations, manages resources and human capital of a functional unit of an organization. The SoMCS branch provides administrative and structural support for faculty members, students, and resources of the school.

MCS staff will have full access to Create/Edit/Delete and view all branch records.

Each "***BRANCH***" table contains the following attributes:

Attribute	Data type	Explanation	Extra Note
BranchID	Integer [PK]	ID of the branch. [3-digit range: 201-299]	Defaults to "201".
Branch	Text	Name of branch [i.e. department or school]	Defaults to "School of MCS"
Faculty	Text	Name of the faculty for the branch	Defaults to "Faculty of Sciences"
HoS	Text	Name of the Head of School with proper title	

4.2.3 Item Branch

ItemBranch is a conceptual entity that facilitates and accounts for storage, management, and tracking of all the stock items and assets for the school. Each stock item must be linked to a stock-holding branch such that ***ItemID/BranchID*** combination forms a ***composite primary key attribute*** and serves as a unique identifier for ITEMBRANCH table.

MCS staff will have full access to Create/Edit/Delete and view all the table records.

Each "***ITEMBRANCH***" table contains the following attributes:

Attribute	Data type	Explanation	Extra Note
ItemID	Integer [PK, FK]	ID of the stock item. [Composite PK with BranchID]	Validate ITEM table. Also serves as a Foreign key.
BranchID	Integer [PK, FK]	ID of the branch. [Composite PK with ItemID]	Defaults to "201". Also serves as a Foreign key
UOM	Text	Unit of Measure	Get from ITEM
Current Stock	Integer	Available Quantity at stock-holding branch	System updates [View only]: Balance increases by PO receipts/Adjust INs Balance decreases by Issues/Adjust OUTs
UnitCost	Decimal	Average cost of a unit item	System updates by PO receipts, or manually maintained
Location	Text	Specific location at MCS	
Reorder Level	Integer	Quantity to compare Current Stock	Triggers replenishment if ***Current Stock*** is Less Than or Equal To ***Reorder Level***

Reorder Quantity	Integer	Suggested Quantity to replenish stock	

4.2.4 Customer

A customer is the final end-user of the stock item. Items are issued to customers from inventory and recorded for reporting purposes. Customers are predominantly staff, students, and MCS operations.

MCS staff will have full access to Create/Edit/Delete and view all customers.

Each "***CUSTOMER***" table contains the following attributes:

Attribute	Data type	Explanation
CustomerID	Integer [PK]	Unique ID of customer [4-digit range: 4001-4999]
Customer	Text	Name of the customer
CustomerType	Text	Allowed values {STAFF, STUDENT, MCS}
Email	Text	Email address of customer

4.2.5 Stock Issue

Stock items used or issued from the inventory are recorded on the ISSUE table. Stock issues are recorded against customer accounts:

- Staff, students, MCS operations

MCS staff will have full access to Create/Edit/Cancel/Delete and view all stock issues records.

Each "***ISSUE***" table contains the following attributes:

Attribute	Data type	Explanation	Extra Note
Issue Number	Integer [PK]	Unique identifier of the issue transaction	4-digit range: 7001-7999
BranchID	Integer [FK]	ID of the issuing branch.	Defaults to '201'
ItemID	Integer [FK]	ID of the item to issue	Validate ITEM table
Quantity	Integer	Quantity issued out of inventory	System updates: *decreases *Current Stock* in ITEMBRANCH by ItemID/BranchID
UOM	Text	Unit of Measure	Get from ITEM [view only]
CustomerID	Integer [FK]	Final end-user of the item	
Purpose	Text	Justification of the issue	

UserID	Text	Login ID of the user	System default
Date	Date	Transaction date	System default

4.2.6 Stock Adjustment

Staff should perform necessary adjustments on stock items to reconcile **Current Stock** balances. Adjustments may be required to correct the balances due to the following reasons:

- Adjust OUT units to account for items deemed damaged, lost, disposed, obsolete, human error, system error
- Adjust IN units to account for donations, new items, extra units found, human error, system error

MCS staff will have full access to Create/Edit/Cancel/Delete and view all stock adjustment records.

Each "***ADJUSTMENT***" table contains the following attributes:

Attribute	**Data type**	**Explanation**	**Extra Note**
Adjustment Number	Integer [PK]	Unique identifier of the adjustment	4-digit range: 5001-5999
Adjustment Type	Text	Stock IN or Stock OUT	
BranchID	Integer [FK]	ID of stock-holding branch.	Defaults to '201'
ItemID	Integer [FK]	ID of item to adjust	Validate ITEM table
Quantity	Integer	Quantity adjusted IN or Adjusted OUT	System updates *Current Stock* in ITEMBRANCH by ItemID/BranchID: *Adjust-IN increases availability *Adjust-OUT decreases availability
UOM	Text	Unit of Measure	Get from ITEM [view only]
Date	Date	Transaction date	
Purpose	Text	Justification for the adjustment	
UserID	Text [FK]	Login ID of the user	System default

4.3 Common Tables

4.3.1 Asset Register

Assets are initially purchased as stock items and later reclassed as assets. Assets are recorded separately on the ASSET register for visibility and reporting purposes.

MCS staff will have full access to Create/Edit/Delete and view all items in the asset register.

Each "***ASSET***" table contains the following attributes:

Attribute	Data type	Explanation
AssetID	Integer [PK]	Unique identifier of the Asset [4-digit range: 3001-3999]
Asset	Text	Name of the asset
Serial Number	Text	SN of the asset
Brand	Text	Brand name
Model	Text	Model name
Date	Date	Purchase date
Unit Cost	Decimal	Purchase cost of the Asset
Location	Text	Physical location at MCS
BranchID	Integer [FK]	ID of the stock-holding branch. Defaults to "201"
Status	Text	Status {new, used, under repair, out-of-service}
PO Number	Integer	PO number of last purchase
ItemID	Integer [FK]	ID of the stock item. Validate ITEM table

4.3.2 User Roles

The system should support multiple users of the system. System access and permission rights are managed at the ROLE level. Each user is assigned a ROLE according to the level of access required by the user. Multiple users with the same access level are linked to a single ROLE.

MCS staff will have full access to Create/Edit/Delete and view all users.

Each "***USERS***" table contains the following attributes:

Attribute	Data type	Explanation
UserID	Text [PK]	Unique ID of the user
User	Text	Name of the user
BranchID	Integer [FK]	ID of the School/Department. Defaults to ''201"
Role	Text	Controls permission and access of the system. [Allowed roles are Administrator, Purchasing Officer, Inventory Officer, EndUser]
Permission rights by ROLE	Administrator – super access Purchasing Officer – full access to Inventory and Purchasing [Create/Edit/Cancel/Delete/View] Inventory Officer – full access to Inventory only [Create/Edit/Cancel/Delete/View] EndUser – View only	

5 Business Rules

The system must enforce these business rules to comply with the school's operational requirements:

5.1 Purchasing

- A supplier may supply many items.
- One GE order may contain many items.
- One purchase order belongs to one supplier.
- Quantity cannot become negative.
- Only approved GE orders may become purchase orders.

5.2 Inventory

- Every asset must have a unique Asset Number.
- Every item must have a unique Item ID
- Quantity cannot become negative.
- Every item on the ItemBranch must exist on the Item table
- Every item on the ItemBranch must be assigned a location
- Every asset item must exist on the ItemBranch table

5.3 Common

- Every user must be linked to a Role
- Every user must exist on the Branch table